THE SPECTRAL POWER OF THE DIGIT FUNCTION

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Abstract. We derive an exact closed-form expression for the spectral power $\Phi(k)$ of the digit partition induced by $\delta(r) = \lfloor br/p \rfloor$ on $\mathbb{Z}/p\mathbb{Z}$. For $k \neq 0$,

$$\Phi(k) = \sum_{d=0}^{b-1} \left| \frac{\sin(\pi k n_d/p)}{\sin(\pi k/p)} \right|^2,$$

where n_d is the size of the d-th digit bin. At $k = 0$, $\Phi(0) = \sum_d n_d^2 = S(p, b)$, recovering the bin-sum from [3]. Parseval's identity gives $\sum_k \Phi(k) = p(p-1)$.

The spectral power captures the magnitude of the bin Fourier coefficients but not their phase. Its zeroth mode $\Phi(0) = S(p, b)$ governs the alignment limit of [3], and a prime p is digit-partitioning in base b if and only if $\Phi(k)$ is constant in k. The cyclic autocorrelation $R(\ell)$ used in [6] to compute the eigenvalue spectrum requires the phase information discarded by squaring, and is treated separately in the sequel via a two-variable cross-spectral function whose diagonal recovers Φ. The present paper identifies the additive Fourier analysis of $\mathbb{Z}/p\mathbb{Z}$ as the natural setting for the digit-bin partition and derives every closed form that follows from magnitude data alone.

1. Introduction

The digit partition of a prime p in base b is the collection of bins $B_d = \{r : 1 \leq r \leq p-1,\ \lfloor br/p \rfloor = d\}$ for $d = 0, 1, \ldots, b-1$. The bin sizes $n_d = |B_d|$ are integer invariants of the pair (p, b). Earlier papers in this series have used the bin sizes through several scalar combinations: the bin-sum $S(p, b) = \sum_d n_d^2$ that governs the alignment limit [3], the alignment α and pairwise alignment σ of the coherence decomposition [4], and the cyclic autocorrelation $R(\ell)$ used in the spectral structure paper [6] to compute the eigenvalues of the cross-alignment matrix [5]. Each of these invariants is some contraction of the bin partition. None of them is a function of a single Fourier mode of the partition itself.

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This paper identifies that single Fourier mode. We define the *spectral power* $\Phi(k)$ of the digit partition as the sum of squared magnitudes of the bin indicators' additive Fourier coefficients, prove that $\Phi(k)$ has a closed form in terms of Dirichlet kernels weighted by the bin sizes, and show that the existing scalar invariants are values or contractions of Φ. The bin-sum $S(p, b)$ is $\Phi(0)$. The digit-partitioning characterization $p \leq b+1$ is equivalent to constancy of $\Phi(k)$ in k. The cyclic autocorrelation $R(\ell)$ is not determined by Φ alone (squaring discards the phase information that R depends on) but is naturally expressed through a two-variable extension of Φ, the cross-spectral function, which is the subject of the sequel paper.

The technical content of this paper is the observation that the bins B_d are contiguous integer intervals (because $\delta(r) = \lfloor br/p \rfloor$ is monotone in r), so each bin indicator's Fourier transform is a geometric sum, and its magnitude is a Dirichlet kernel. Once that is in place, the closed form for $\Phi(k)$ follows in one line from the definition. The recovery of $S(p, b)$ as $\Phi(0)$ is a corollary of the same identity, and Parseval's identity gives the total spectral energy $\sum_k \Phi(k) = p(p-1)$ as a sanity check.

2. The Bin Partition and Its Fourier Expansion

Definition 1. For a prime p not dividing b, the *bin indicator* of digit d is the function $\mathbf{1}_{B_d} : \{1, \ldots, p-1\} \to \{0, 1\}$ defined by

$$\mathbf{1}_{B_d}(r) = \begin{cases} 1 & \text{if } \lfloor br/p \rfloor = d, \\ 0 & \text{otherwise.} \end{cases}$$

The bin B_d is a contiguous interval of integers in $\{1, \ldots, p-1\}$ of length n_d.

Since B_d is a contiguous interval, the additive Fourier transform of $\mathbf{1}_{B_d}$ is a geometric sum.

Lemma 2. *The Fourier coefficient of $\mathbf{1}_{B_d}$ at frequency k is*

$$\widehat{\mathbf{1}}_{B_d}(k) = \sum_{r \in B_d} e^{-2\pi i k r/p}.$$

For $k \neq 0$, this equals a Dirichlet kernel evaluation:

$$|\widehat{\mathbf{1}}_{B_d}(k)| = \left| \frac{\sin(\pi k n_d/p)}{\sin(\pi k/p)} \right|.$$

For $k = 0$, $|\widehat{\mathbf{1}}_{B_d}(0)| = n_d$.

Proof. The bin B_d is an interval of n_d consecutive integers starting at some a_d. The sum is geometric:

$$\widehat{\mathbf{1}}_{B_d}(k) = e^{-2\pi i k a_d/p} \cdot \frac{1 - e^{-2\pi i k n_d/p}}{1 - e^{-2\pi i k/p}}.$$

Taking absolute values cancels the phase and gives the stated Dirichlet kernel form. □

3. The Spectral Power

Definition 3. The *spectral power* of the digit function at frequency k is

$$\Phi(k) = \sum_{d=0}^{b-1} |\widehat{\mathbf{1}}_{B_d}(k)|^2.$$

Theorem 4. *The spectral power has the exact closed form*

$$\Phi(k) = \sum_{d=0}^{b-1} \left| \frac{\sin(\pi k n_d/p)}{\sin(\pi k/p)} \right|^2$$

for $k \neq 0$, and $\Phi(0) = \sum_d n_d^2 = S(p, b)$.

Proof. Immediate from Lemma 2 and the definition. □

Corollary 5. $\sum_{k=0}^{p-1} \Phi(k) = p(p-1)$.

Proof. By Parseval's identity applied to each bin indicator: $\sum_k |\widehat{\mathbf{1}}_{B_d}(k)|^2 = p \cdot n_d$. Summing over d: $\sum_k \Phi(k) = p \sum_d n_d = p(p-1)$. □

4. Worked Example at $p = 13$ in Base 10

To make Theorem 4 concrete, we compute Φ explicitly at $p = 13$ in base 10. The bin sizes $n_d = |\{r \in \{1, \ldots, 12\} : \lfloor 10r/13 \rfloor = d\}|$ are

$$(n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7, n_8, n_9) = (1, 1, 1, 2, 1, 1, 2, 1, 1, 1).$$

Eight singleton bins and two doubled bins, summing to $\sum_d n_d = 12 = p - 1$. The bin-sum is

$$\Phi(0) = S(13, 10) = 8 \cdot 1 + 2 \cdot 4 = 16.$$

For $k \neq 0$, the singleton bins contribute Dirichlet kernel value 1, and the doubled bins contribute $|2\cos(\pi k/13)|^2 = 4\cos^2(\pi k/13)$. Summing,

$$\Phi(k) = 8 + 8\cos^2\left(\frac{\pi k}{13}\right), \qquad k = 1, 2, \ldots, 12.$$

The total spectral energy is $\sum_{k=0}^{12} \Phi(k) = 16 + \sum_{k=1}^{12} [8 + 8\cos^2(\pi k/13)] = 16 + 96 + 8 \cdot 11/2 = 16 + 96 + 44 = 156$, in agreement with Corollary 5

since $p(p-1) = 13 \cdot 12 = 156$. The constant $11/2$ in the cosine sum comes from the standard identity $\sum_{k=1}^{p-1} \cos^2(\pi k/p) = (p-2)/2$.

5. Equivalent Characterization of Digit-Partitioning Primes

The spectral power gives a clean spectral characterization of the digit-partitioning property.

Proposition 6. *A prime p is digit-partitioning in base b (equivalently, $p \le b+1$) if and only if $\Phi(k)$ is constant in k for $k \in \{0, 1, \ldots, p-1\}$.*

Proof. If p is digit-partitioning then $n_d \le 1$ for every d, so each Dirichlet kernel $|\sin(\pi k n_d/p)/\sin(\pi k/p)|$ is identically 1 at nonzero k, and $\Phi(k) = p-1$ for $k \neq 0$. Also $\Phi(0) = \sum_d n_d^2 = p-1$ since each nonzero n_d equals 1. So Φ is constant.

Conversely, suppose Φ is constant. Then $\Phi(0) = \sum n_d^2$ equals the common value $\Phi(k)$ for any nonzero k. By Corollary 5, the constant value is $\Phi(k) = p-1$ for all k. So $\sum_d n_d^2 = p-1 = \sum_d n_d$, which forces $n_d \in \{0, 1\}$ for every d (since $n_d^2 = n_d$ only at these values). Hence each nonempty bin has exactly one element, so the bin partition is a digit partition, which by [2] is equivalent to $p \le b+1$. □

6. The Position of Φ in the Chain

The spectral power is the magnitude-only data of the bin Fourier coefficients. To recover the cyclic autocorrelation $R(\ell)$ used in [6] for the eigenvalue spectrum, the phase information discarded by squaring is needed. The "same digit" indicator expands as

$$\mathbf{1}[\delta(r) = \delta(s)] = \sum_d \mathbf{1}_{B_d}(r)\, \mathbf{1}_{B_d}(s),$$

and substituting $s = b^\ell r \bmod p$ followed by the inverse Fourier expansion of each indicator gives

$$R(\ell) = \frac{1}{p} \sum_{k=0}^{p-1} \sum_d \widehat{\mathbf{1}}_{B_d}(k)\, \overline{\widehat{\mathbf{1}}_{B_d}(b^{-\ell} k)}.$$

The inner sum at fixed ℓ involves cross-frequency products $\widehat{\mathbf{1}}_{B_d}(k)\, \overline{\widehat{\mathbf{1}}_{B_d}(b^{-\ell} k)}$, which are not captured by $|\widehat{\mathbf{1}}_{B_d}(k)|^2 = \Phi(k)$ alone. The two-variable cross-spectral function $G(k, k') = \sum_d \widehat{\mathbf{1}}_{B_d}(k)\, \widehat{\mathbf{1}}_{B_d}(k')$, of which Φ is the antidiagonal $G(k, -k)$, is the natural object that contains the phase data; it is introduced in the sequel paper, where the closed-form expression for $R(\ell)$ as a sum of G along an orbit path is proved.

7. Unification of Earlier Invariants

Several scalar invariants of the alignment program are contractions of Φ. The alignment limit of [3] is governed by the zeroth mode $\Phi(0) = S(p,b)$, since the limiting alignment $\mathcal{L}(p,b) = (p-1+S(p,b))/(p(p-1))$ in the equidistributed case is a function of $\Phi(0)$ alone. The coherence decomposition $\alpha = F + \sigma$ of [4] expresses α, σ, and F through $\Phi(0)$ together with the orbit structure of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$. The digit-partitioning characterization, by Proposition 6, is the statement that Φ is constant in k. The eigenvalue spectrum of [6] is the DFT of the cyclic autocorrelation $R(\ell)$, which is not determined by Φ alone but which is naturally expressed through the cross-spectral function whose diagonal is Φ.

In each case, Φ encodes the interaction between the Euclidean geometry of the digit function (the bin sizes n_d, which are integer counts of consecutive residues) and the additive harmonic analysis of $\mathbb{Z}/p\mathbb{Z}$ (the Dirichlet kernel at each frequency). It is the simplest single object from which the magnitude-side invariants of the program can be read.

8. Remarks

The Dirichlet kernel and the digit function. The Dirichlet kernel $D_N(x) = \sin(N\pi x)/\sin(\pi x)$, a classical object in harmonic analysis [7, 9], appears because the digit bins are contiguous intervals. This is a consequence of the monotonicity of the digit function $\delta(r) = \lfloor br/p \rfloor$: the function is nondecreasing in r, and each preimage B_d is a contiguous interval (possibly empty). The spectral power inherits the Dirichlet kernel's properties: concentration at low frequencies for large bins, oscillatory behavior at high frequencies.

Additive vs multiplicative characters. The Legendre symbol (a multiplicative character) was tested in the spectral structure paper [6] and found insufficient to determine $R(\ell)$. The spectral power uses additive characters instead, which succeed because the bin partition is defined by an additive condition (contiguous intervals in $\mathbb{Z}/p\mathbb{Z}$). The digit function bridges multiplicative structure (the orbit of b) with additive structure (the bin intervals), and the spectral power is the natural invariant of this bridge. The interplay between additive and multiplicative characters in number-theoretic contexts is treated systematically in [9].

Toward L-functions. The spectral power $\Phi(k)$ is defined on $\mathbb{Z}/p\mathbb{Z}$ and is symmetric ($\Phi(k) = \Phi(p-k)$). Its values at specific k can be related to Ramanujan sums [8] and Kloosterman-type expressions. Whether the

generating function $\sum_k \Phi(k) z^k$ or the Dirichlet series $\sum_p \Phi_p(k) p^{-s}$ (summing over primes at a fixed frequency k) exhibit analytic continuation or functional equations is an open question.

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